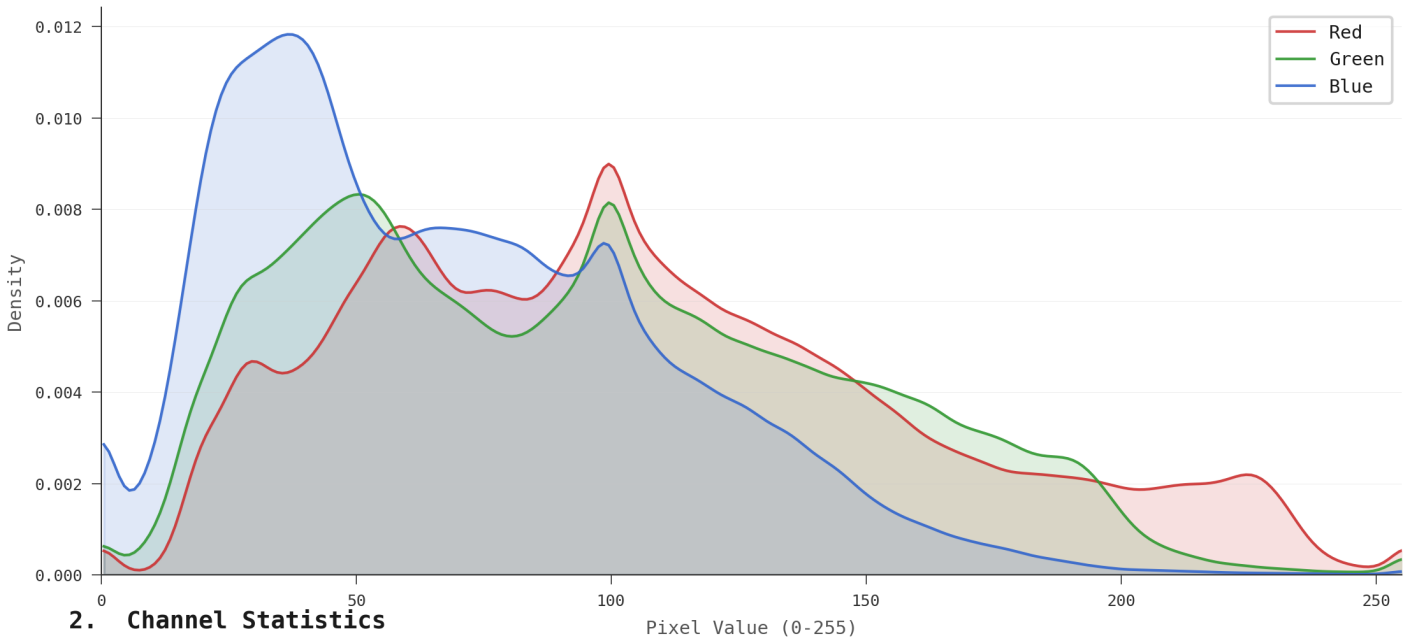
Reference Data Sheet: KODIM14

Subject: Whitewater Rafting Group

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	105.7633	54.3565	2954.6247	-0.4082	0.5211	0	255
Green	94.2856	50.9241	2593.2623	-0.7031	0.3936	0	255
Blue	68.6271	41.0520	1685.2690	-0.0027	0.6833	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8613	0.4568
G	0.8613	1.0000	0.7107
B	0.4568	0.7107	1.0000

Pairwise Pearson Correlation Coefficients

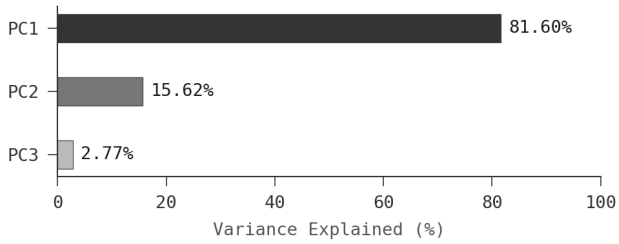
r(R-G)	=	**0.861318**
r(R-B)	=	**0.456785**
r(G-B)	=	**0.710744**
Avg |r|	=	**0.676283**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2954.6322} & 2384.1802 & 1019.2930 \\
2384.1802 & \mathbf{2593.2689} & 1485.8409 \\
1019.2930 & 1485.8409 & \mathbf{1685.2732}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM14

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.6571**	-0.6470	-0.3868	5902.5075
PC2	-0.5608	0.0768	0.8244	1130.1467
PC3	0.5037	-0.7586	0.4133	200.5202

6. Derived Metrics

Condition Number

29.44

(l1/l3) PC1/PC2: 5.22 | PC2/PC3: 5.64

Blue Channel Independence

47.6%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.6571 | Red dominant

Classification: Two-Dimensional (PC1 = 81.60%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.961580	0.942926	0.952253
Green	0.959062	0.938324	0.948693
Blue	0.945840	0.914840	0.930340

8. Average Local Variance (3x3)

Channel	Value
Red	228.0342
Green	209.8691
Blue	177.3103

9. Redundancy Profile

Classification: Two-Dimensional | Condition Number: 29.44 | PC1: 81.60%

Correlation: R–G pair leads at r = 0.861318 | G–B: 0.710744 | R–B: 0.456785

Blue Channel: 47.6% independent | 3D color; major blue variance

Spatial: Avg autocorrelation: 0.943762 | Avg local variance: 205.0712

Eigenvector: Red dominant | R (-0.6571) leads | G (-0.6470), B (-0.3868)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."